

Cyclone Heat Potential

Daily TCHP, D26 and ocean heat content over the Bay of Bengal and Arabian Sea, from satellites alone.

PORT	PAGE FILE	MODULE	UNITS
8509	app/phase2/cyclone_heat_page.py	phase2/derived/heat_content.py	kJ/cm2, metres, GJ/m2

This is the feature that makes the project nationally relevant rather than merely accurate. Sea-surface temperature alone does not tell you whether a cyclone will intensify; the depth of the warm water underneath it does. That quantity is what this page produces, every day, without a single in-situ float.

1. What this feature is, in plain words

A cyclone does not just take heat from the surface. Its own winds churn the sea, dragging up cooler water from below. If the warm layer is thin, the storm quickly pulls up cold water and cools its own fuel supply. If the warm layer is deep, the churning brings up more warm water and the storm keeps intensifying.

Tropical Cyclone Heat Potential (TCHP) measures exactly that: the heat stored above the 26 degrees Celsius isotherm - 26 being the accepted threshold below which tropical cyclones do not sustain themselves. **D26** is the depth of that isotherm, meaning how thick the warm layer is. **Ocean heat content** is the total heat down to a chosen depth.

Until now, producing TCHP over this basin needed subsurface observations - floats, or a reanalysis that runs days behind. We produce it daily, from satellite inputs, everywhere in the basin, because the model underneath supplies the full temperature column.

Every value also carries an uncertainty, propagated through the integral by sampling. And the page states, on screen, that this uncertainty is a **lower bound** - the model provides per-depth variance with no cross-depth correlation, and adjacent depths are physically likely to be correlated, so the true spread is wider than the number shown.

THE EVERYDAY PICTURE

Surface temperature tells you the fuel tank is warm. TCHP tells you how deep the tank is. A storm crossing a thin warm layer runs out of fuel; one crossing a deep warm layer does not - and that is often the difference between a storm and a disaster.

2. What you actually see on the screen

Three related products, one date, one basin:

- A daily **TCHP** map in kJ/cm2 across the Bay of Bengal and the Arabian Sea.
- A **D26** map - the depth of the 26 degree isotherm, computed by linear interpolation between the last warm level and the first cold one.
- An **ocean heat content** map to a chosen reference depth.
- A point inspector giving all three values with a plus-or-minus 1 sigma range beside each.
- Land drawn as blank rather than as zero - because a zero would read as 'cold', which is a false claim about a place with no water at all.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/cyclone_heat_page.py --server.port 8509
```

- 1 Run the command and open **http://localhost:8509**.
- 2 Pick a date in the pre-monsoon or post-monsoon cyclone season - October and November are the Bay of Bengal's dangerous months.
- 3 Show the TCHP map first. Point at the deep-warm-pool regions and say plainly: a storm crossing there will find fuel; one crossing the thin areas will not.
- 4 Switch to D26 and explain that this is the thickness of that fuel layer in metres.
- 5 Use the point inspector on a specific location and read the plus-or-minus 1 sigma range aloud, then say it is a lower bound and why.
- 6 To produce the same fields as files for another system, use the export script below.

WRITE THE FIELDS TO NETCDF FOR ONE DATE OR A RANGE

```
PYTHONPATH=src python scripts/phase2/make_heat_content.py --date 2025-10-02
```

4. Why the problem statement needs this

The problem statement sits under a disaster-management theme with INCOIS as the sponsor. TCHP is the single most directly operational quantity this project can produce: it is what cyclone-intensity forecasters actually consume, and India's cyclone exposure is among the highest in the world.

It also demonstrates that a temperature-only model is genuinely useful. TCHP needs no salinity, so it lies entirely inside our compliance boundary - it is a satellite-derived product with no reanalysis smuggled in, which is exactly why Feature 5 refuses the fields that would need salinity while this one proceeds.

PS requirement	How this feature answers it
Daily, basin-wide output	TCHP, D26 and OHC for every ocean cell, every day in the bundle.
Satellite inputs only	Derived from the shipped temperature field, which needs no salinity - so nothing reanalysis-based enters the product.
Useful over the Bay of Bengal and Arabian Sea	Both basins covered, and their seasonal heat structures are visibly different.
Uncertainty reported	Model per-depth sigma propagated through the integral by sampling, labelled as a lower bound.

5. The measured numbers behind it

Constants, conventions and guards - all chosen explicitly and tested [VERIFIED]:

Choice	Value	Why it is stated rather than assumed
TCHP constants	$c_p = 4000 \text{ J/(kg degC)}$, $\rho = 1026 \text{ kg/m}^3$	The TCHP-literature convention (Leipper and Volgenau 1972). Our climate-budget module uses 3985 / 1025 - they differ by ~0.4% and are kept as each field's own convention, not silently unified.
Realistic range	about 20-120 kJ/cm ² for a tropical column	A test guards this range explicitly. A result near 1,000,000 would be a unit bug, and the unit conversion is a factor of 1e-7.
Surface colder than 26 degC	TCHP = 0	There is genuinely no warm layer. Zero is the correct answer here.
Land or an all-missing column	NaN, never 0	A zero would read as 'cold water'. An absence is not a value.
A warm layer below a cold one	excluded	The warm layer is the surface-connected one, so a deep re-warming is correctly left out rather than counted as fuel.

- The whole-basin field and the single-point calculation call the **same** unit-tested function, and a test asserts field value equals scalar value at each cell. There is no second implementation of TCHP that could drift from the first.

6. The impact - and who it is for

Who	What changes for them
IMD / INCOIS cyclone forecasters	The subsurface intensification variable, daily and basin-wide, without waiting for a float to surface.
Coastal district administrations	Earlier and better-founded signals about whether an approaching system has fuel to intensify.
Disaster-response planners	A physical basis for evacuation timing, which is the decision that saves the most lives.
Fishers and port operators	Advance warning tied to the ocean's actual energy state rather than to surface temperature alone.

WHY SOCIETY GAINS FROM THIS

The Bay of Bengal produces a small share of the world's tropical cyclones and a very large share of its cyclone deaths, because the storms meet a low-lying, densely populated coast. Any improvement in intensification forecasting converts directly into evacuation lead time. This feature is the project's clearest line from a satellite pixel to a life saved.

7. Say this to the jury

Surface temperature does not tell you whether a cyclone will intensify - the depth of the warm water underneath does, because the storm's own winds drag up whatever is below. That quantity is Tropical Cyclone Heat Potential, and producing it has always required subsurface observations. We produce it daily, everywhere in the basin, from satellites alone. And every value carries an uncertainty which we label a lower bound, because our model gives per-depth variance without cross-depth correlation and we will not present that as if it were the full story.

This is the feature to lead with if the jury asks 'why does this matter to India'.

8. Questions the jury will ask

Q. Is TCHP not already available operationally?

A. It is, from products built on floats and reanalysis, which are sparse in space or lag in time. Our contribution is producing it from satellite inputs, daily, at every one of 24,000 grid cells in this basin - and being explicit that our

temperature error propagates into it.

Q. Why is your uncertainty a lower bound rather than the real one?

A. Because the model emits a variance per depth with no covariance between depths, so our sampling draws each depth independently. Real errors at adjacent depths are almost certainly correlated, which would widen the spread. We state that on screen rather than presenting the narrower number as complete.

Q. What happens over land or on the shelf?

A. Land returns NaN, never zero, because zero would read as cold water. A column that does not reach the reference depth returns NaN rather than a partial integral labelled as a full one - that exact failure occurred once in Phase 1 and was fixed.

9. The honest limit - say it before they find it

ITS ACCURACY IS INHERITED FROM THE TEMPERATURE FIELD

TCHP is an integral of our reconstructed temperature, so it carries our temperature error - including the +0.1003 degC warm bias, which will tend to overstate heat content slightly. Integrals are far more tolerant of bias than threshold crossings are, which is why the equivalent stage-2 heat content survived validation (bias -0.024 GJ/m2) while the mixed-layer depth from the same model did not. The uncertainty shown is a measured lower bound, not the true uncertainty.

IN ONE LINE

The cyclone-intensification variable - heat above the 26 degree isotherm - produced daily from satellites alone, with its uncertainty labelled a lower bound.

26 degC

THE ISOTHERM THAT DEFINES THE FUEL
LAYER

20-120

KJ/CM2, THE GUARDED REALISTIC RANGE

NaN

OVER LAND - NEVER A ZERO THAT READS AS
COLD

Feature 9 of 17. Next: Feature 10.